CLINICAL · EAR

Chronic suppurative otitis media without cholesteatoma

Chronic suppurative otitis media of the mucosal, or safe, type is a wet ear discharging through a hole that will not heal. It rarely threatens life, but it wears down hearing and patience. The aim is a dry, safe ear: clean it properly, treat it topically for long enough, prove there is no cholesteatoma, and then decide whether repair is worth it.

AT A GLANCE

- A permanent central perforation with recurring mucoid discharge.
- Clean it first, then topical quinolone for weeks.
- Keep water out and treat the nose and throat.
- Prove there's no cholesteatoma.
- Repair the drum once dry.

1 CHAPTER ONE Overview

A persistent hole and a discharging ear.

1.1 Definition

Chronic suppurative otitis media is long-standing infection of part or all of the middle ear cleft, with ear discharge and a permanent perforation. It is classically divided into two types: the *tubotympanic* or *safe* type, which involves the mucosa and the anteroinferior cleft, and the *atticoantral* or *unsafe* type, which involves bone and usually cholesteatoma. This chapter covers the safe, mucosal type.

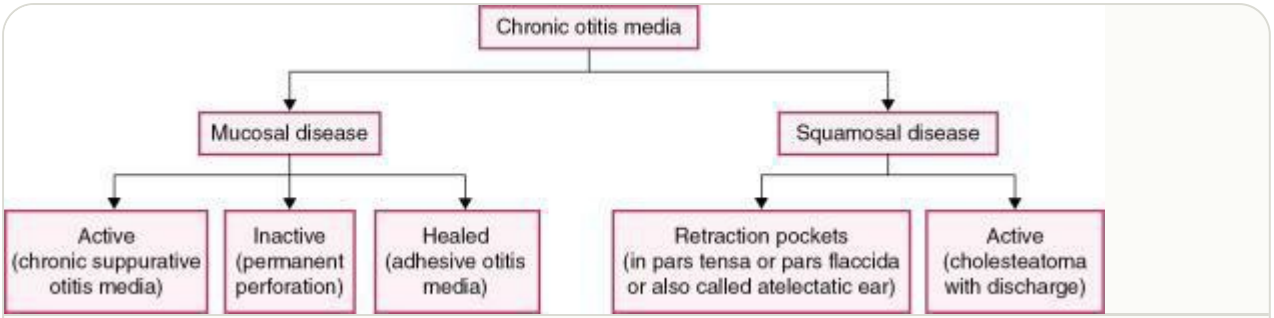


FIGURE 1 Classification of chronic otitis media into tubotympanic (mucosal) and atticoantral (squamous) types.

Dhingra, fig. 11.7

1.2 Clinical stages

1.2.1 Active mucosal disease

Discharge through a pars tensa perforation.

1.2.2 Inactive mucosal disease

A dry, permanent perforation that may become active with a cold or water.

1.2.3 Healed chronic otitis media

An atrophic, healed drum, often with tympanosclerosis and some conductive loss.

1.3 Why it matters

It is a common cause of preventable hearing loss worldwide, especially where crowding, poverty and limited access to care allow acute infections to go untreated.

2 CHAPTER TWO Pathogenesis

Why the hole does not heal and the ear keeps running.

2.1 Origins

Most cases follow repeated acute otitis media in childhood, or a perforation from acute necrotising otitis or trauma. Chronic effusion and Eustachian tube

dysfunction keep the mucosa inflamed.

2.2 The permanent perforation

Once the squamous epithelium of the drum grows round the edge of the hole and meets the mucosa on the inside, the perforation behaves like a lined fistula and will not close on its own.

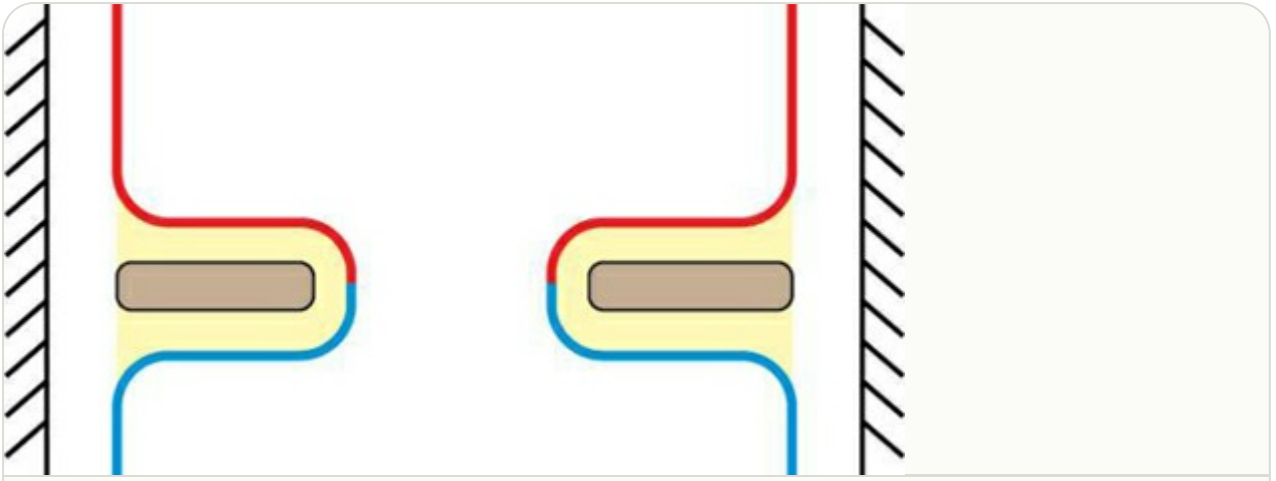


FIGURE 2 A permanent perforation of the pars tensa: the epidermal and mucosal layers meet across its edge, so it behaves like a non-healing fistula.
Dhingra, fig. 11.5

2.3 Why the discharge recurs

2.3.1 Reinfection through the tube

Infection ascends from the nose, sinuses, adenoids and tonsils, especially in children, and nose blowing pushes secretions into the middle ear.

2.3.2 Reinfection through the canal

Water entering through the perforation during bathing or swimming.

2.3.3 Biofilms

Bacteria, particularly *Pseudomonas aeruginosa*, form biofilms on the mucosa that shield them from antibiotics and immune cells, explaining the relapses after apparently successful treatment.

2.4 Organisms

Pseudomonas aeruginosa, *Proteus*, *Escherichia coli* and *Staphylococcus aureus* are the commonest aerobes, often with anaerobes such as *Bacteroides*. Fungi may follow repeated antibiotic drops.

3 CHAPTER THREE Clinical features

The safe ear.

3.1 Symptoms

3.1.1 Discharge

Profuse, mucoid or mucopurulent, without a foul smell, and intermittent: it flares with colds and with water in the ear and dries between episodes.

3.1.2 Hearing loss

Conductive, of variable degree, and rarely over 50 dB. Some patients hear better when the ear is wet, because the discharge shields the round window and restores a phase difference between the windows.

3.2 Signs

3.2.1 The perforation

Central, with a rim of drum all round it. It may lie anterior, posterior or inferior to the handle of malleus, or be subtotal. A large perforation may expose the ossicles and the promontory.

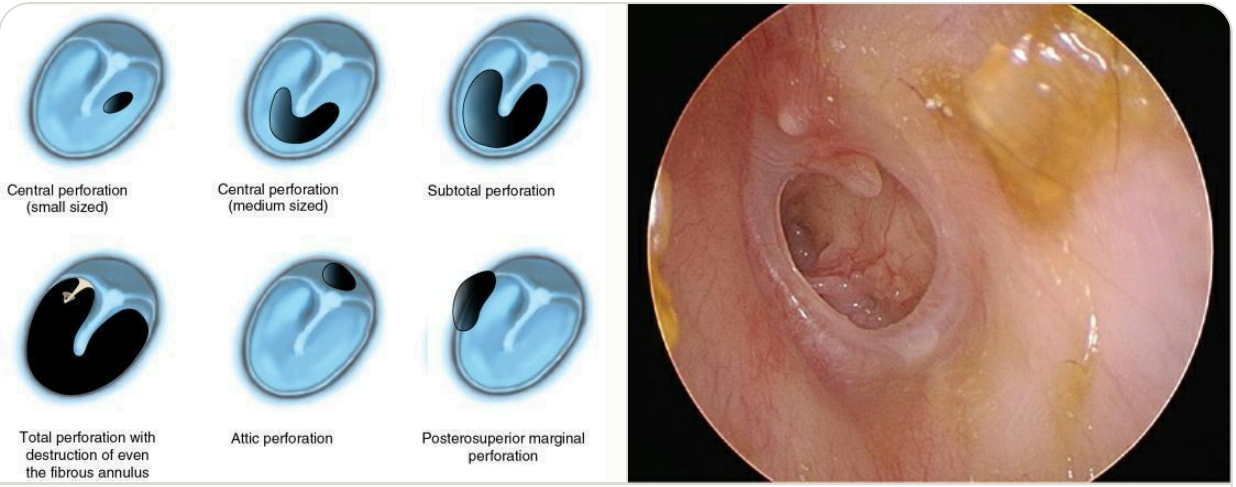


FIGURE 3 First, the types of perforation seen in the pars tensa. Second, a large central perforation.

Dhingra, figs 11.10 and 11.11

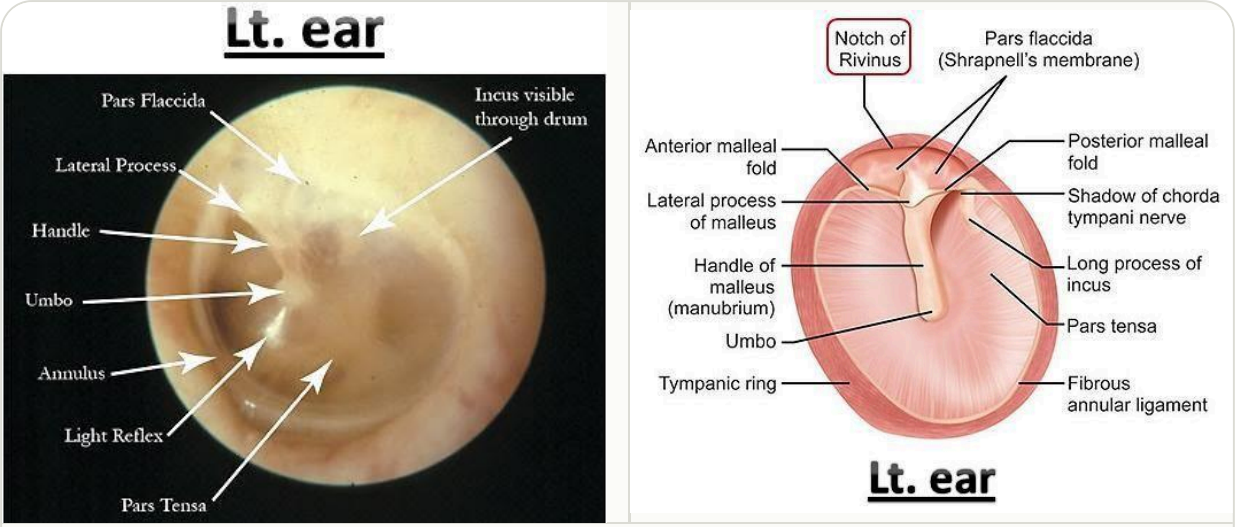


FIGURE 4 A normal tympanic membrane and its parts, for describing the site of a perforation.

Anatomy atlas · ENT Steps p. 36

3.2.2 The middle ear mucosa

Normal and pale when inactive, red and oedematous when active. A pale polyp may fill the perforation.

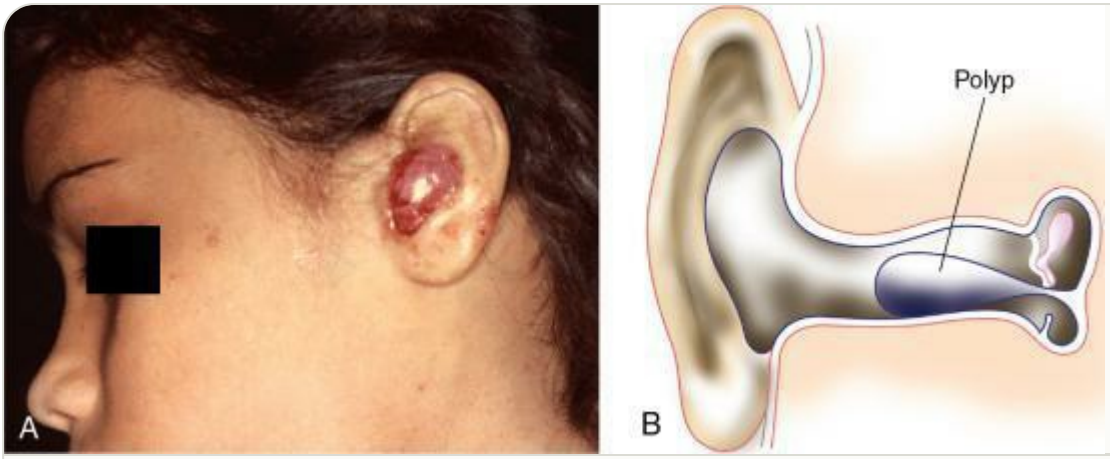


FIGURE 5 An aural polyp in the canal, and how a polyp from the promontory can present through the perforation.

Dhingra, fig. 11.9

NEVER AVULSE A POLYP

A polyp may arise from the stapes, the facial nerve or the lateral semicircular canal. Pulling it out risks facial palsy or a dead ear; remove it under the microscope or at surgery, and send it for histology.

3.3 Safe or unsafe?

MUCOSAL VERSUS SQUAMOUS CHRONIC OTITIS MEDIA

	MUCOSAL (SAFE)	SQUAMOUS (UNSAFE)
Discharge	Profuse, mucoid, not foul	Scanty, foul-smelling
Perforation	Central	Attic or posterosuperior marginal
Polyp	Pale	Red, fleshy
Cholesteatoma	Absent	Present
Complications	Rare	More common
Hearing	Mild to moderate conductive	Conductive or mixed

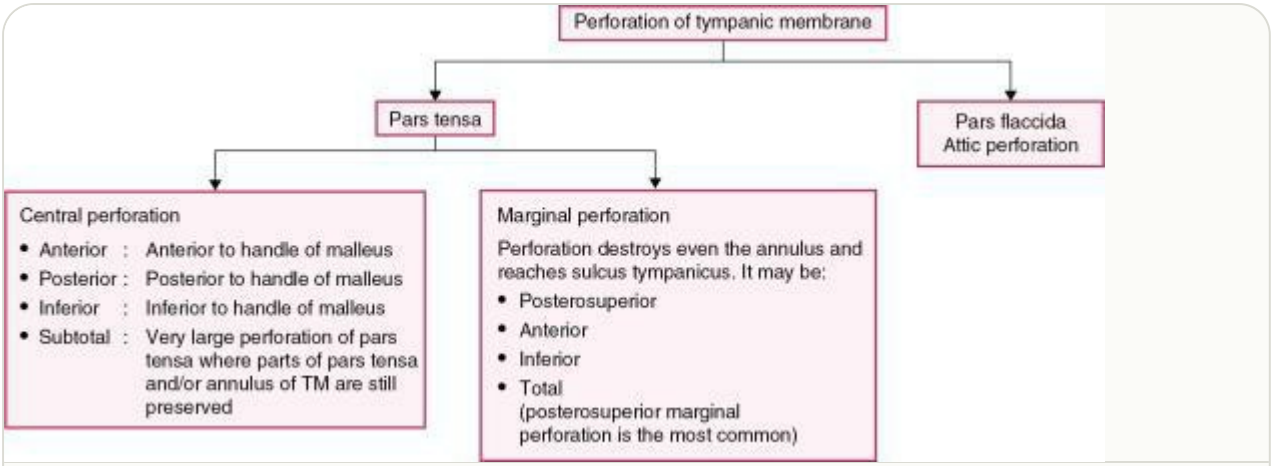


FIGURE 6 Sites of perforation: central perforations belong to the safe type; attic and posterosuperior marginal perforations to the dangerous type.

Dhingra, fig. 11.8

4 CHAPTER FOUR Investigations

Look, test, and image when needed.

4.1 Examination under the microscope

Every chronic ear should be cleaned and examined with the microscope. An apparently dry ear can hide discharge, and rarely cholesteatoma coexists with a central perforation.



FIGURE 7 Examination of the ear under the operating microscope.
Dhingra, fig. 11.12

4.2 Swab for culture

Guides the choice of drops when the ear does not respond.

4.3 Tuning forks and audiogram

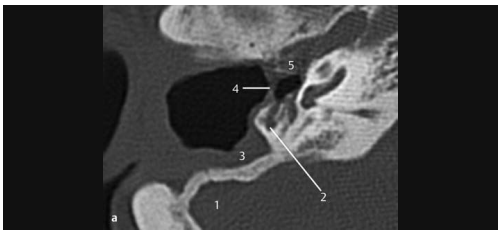
Measure the conductive loss and bone conduction. An air-bone gap over about 40 dB suggests ossicular discontinuity, often erosion of the long process of the incus.

4.4 Imaging

CT is not routine for mucosal disease, where it shows no bone erosion. It is used when cholesteatoma or a complication is suspected, and before revision surgery. MRI is reserved for intracranial concerns.

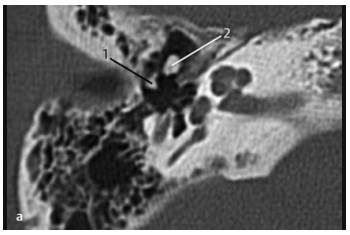
IMAGING IN THE APP 3 cases

A discharging modified radical cavity, and ossicular luxation and fixation.



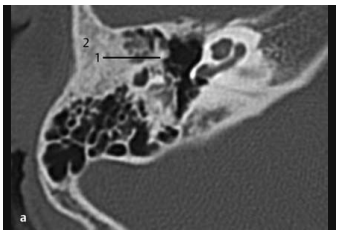
Modified Radical Cavity, Persistent Infections

After surgery · 2 images



Luxation of the Ossicular Chain

Trauma · 10 images



Fixation of the Ossicular Chain

Bone disease · 3 images

4.5 Biopsy

For granulation that persists despite treatment, to exclude tuberculosis and malignancy.

4.6 Differential diagnosis

- **Cholesteatoma:** scanty foul discharge, attic or marginal defect, white flakes.
- **Tuberculous otitis media:** painless thin discharge, pale granulation, multiple perforations, early facial palsy.
- **Granular myringitis:** granulation on the drum itself, without a perforation.
- **Carcinoma of the temporal bone:** pain, bleeding, and granulation that does not settle.

5 CHAPTER FIVE Management

Dry the ear, then decide on repair.

5.1 Aural toilet

Drops cannot work through pus. Clean the ear under the microscope by suction, dry mopping or gentle saline irrigation; never use forceful syringing.

5.2 Topical treatment

5.2.1 Drops

Quinolone drops are preferred when the drum is perforated. Continue for four to six weeks after debridement when the ear is persistently active.

5.2.2 Technique

Tragal pumping after instilling the drops pushes them through the perforation into the middle ear.

5.2.3 Acidifying drops

Acetic acid 1.5% helps against *Pseudomonas*.

WHERE THE BOOKS DIFFER · OTOTOXIC DROPS

Dhingra	Lee
Advises avoiding aminoglycoside and polymyxin drops with a perforation.	Lists neomycin and polymyxin preparations among the options.

Quinolone drops are the safer choice with an open middle ear.

5.3 Systemic antibiotics

Mainly for acute flares, chosen by culture.

5.4 Treating the source

Treat infected tonsils, adenoids and sinuses, and allergy; avoid hard nose blowing.

5.5 Keep water out

Cotton wool with petroleum jelly, or a moulded plug, during bathing; no swimming while the ear is active.

5.6 Surgery

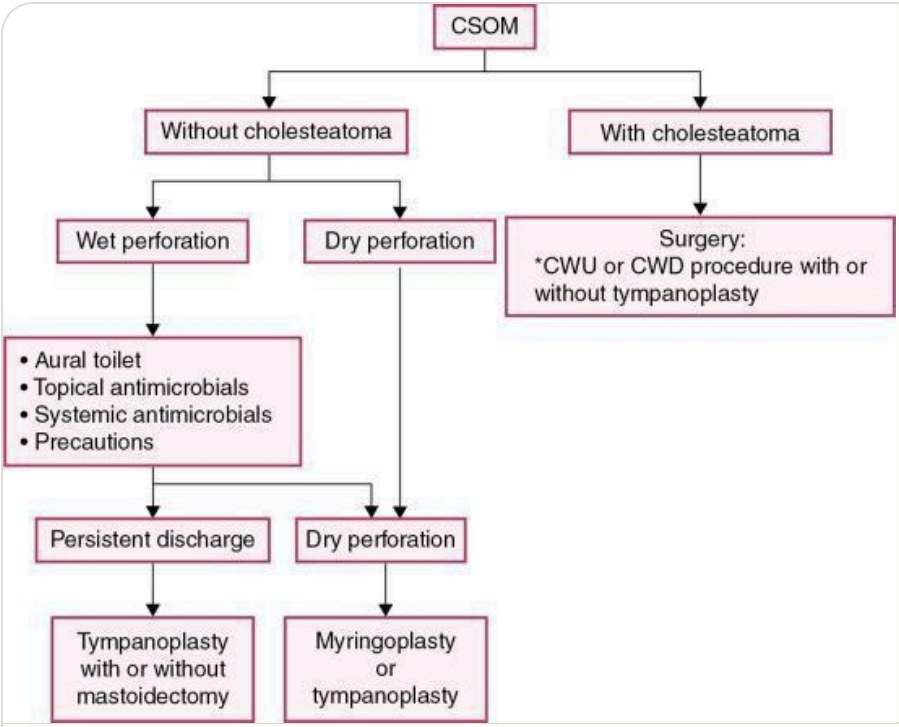


FIGURE 8 A management pathway for chronic suppurative otitis media, from medical treatment to tympanoplasty and mastoid surgery.

Dhingra, fig. 11.14

5.6.1 Tympanoplasty

Once the ear has been dry for some weeks, repair of the perforation (myringoplasty), with ossicular reconstruction if needed, restores a closed, protected middle ear. Success is 60 to 90%, higher when air passes through the perforation on Valsalva, a sign of a working tube.

5.6.2 Mastoidectomy

For discharge that persists despite proper medical treatment. Cortical mastoidectomy may be combined with tympanoplasty; on its own it does not improve the rate of drum closure.

5.6.3 Hearing aids

For patients unfit for or unwilling to have surgery, ideally once the ear is dry.

PEARLS

- The other ear predicts how well the Eustachian tube works.

- Think of tuberculosis when an ear fails to respond, with pale granulation, multiple holes and facial weakness.
- No treatment of the Eustachian tube has proven benefit.

OPERATIVE GUIDES IN THEATRE 3 operations

Myringoplasty and Tympanoplasty
Myringoplasty or tympanoplasty

Ossicular Chain Reconstruction
Ossicular reconstruction

Mastoid Surgery
Tympanomastoidectomy for refractory disease

CHAPTER SIX

6 Complications

Uncommon in the safe ear, but real.

6.1 Ossicular erosion

The long process of the incus is a vascular watershed and erodes first, giving a larger conductive loss.

6.2 Sensorineural hearing loss

Toxins crossing the round window can damage the basal turn of the cochlea, producing a high-frequency loss.

6.3 Tympanosclerosis

Chalky hyaline plaques in the drum and middle ear, which may fix the ossicles.

6.4 Others

Rarely, mastoiditis, facial palsy, labyrinthitis or intracranial infection in an acute flare.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Safe versus unsafe:** the six-row comparison.
- **Why the perforation does not heal,** and why the discharge recurs.
- **Aural toilet,** quinolone drops and water precautions.
- **Hearing better when the ear is wet.**
- **Never avulse a polyp.**
- **Indications for tympanoplasty,** and what predicts success.

7.2 Common pitfalls

COMMON PITFALLS

- Prescribing drops into an ear full of pus.
- Using aminoglycoside drops through a perforation without need.
- Calling a chronic ear safe without examining it under the microscope.
- Missing tuberculosis or malignancy in granulation that will not settle.

Written for this app from Oxford Textbook of Otolaryngology ch. 8, 21, Oxford Specialist Handbook ch. 16, Oxford Handbook of ENT ch. 6, Dhingra ch. 11 and K. J. Lee ch. 22. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

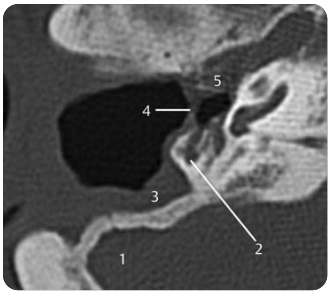
On imaging

The imaging of this topic: key findings, differentials and representative scans.

MIDDLE EAR & MASTOID · AFTER SURGERY

Modified Radical Cavity, Persistent Infections

- Attention must be paid to the above-mentioned presence of residual cells. Residual cholesteatoma may present as smooth-bordered, rounded swellings covered by an epithelial or mucosal lining. In cases of destruction of the bony outline to the middle of posterior cranial fossa, a complementary MRI might be helpful for evaluating the extent of the pathology and its relation to intracranial structures.
- DWI is especially helpful in discriminating between recurrent or residual cholesteatoma and granulation tissue.



Persistent secretion from a modified radical cavity.

Dunnebier, Imaging for Otolaryngologists, Fig. 3.25 a

MIDDLE EAR & MASTOID · TRAUMA

Luxation of the Ossicular Chain

IMAGING FINDINGS

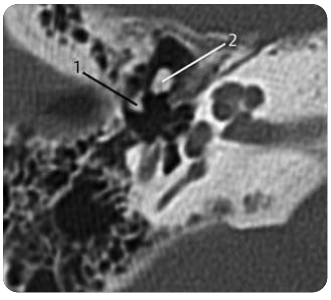
- Separation of the incudostapedial joint; incus rotated or displaced
- Loss of the "ice-cream cone" between malleus head and incus body
- Associated temporal bone fracture through the canal and middle ear

TOP DIFFERENTIALS

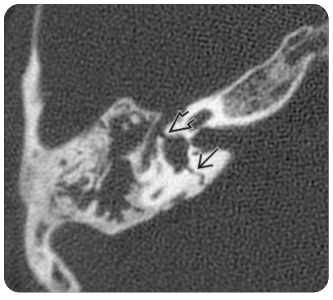
- Ossicular erosion from chronic otitis
- Congenital ossicular malformation
- Partial volume on a single slice

CHECKLIST

- Compare with the other ear on the same slice
- Check facial canal, tegmen and otic capsule



CT, axial. Traumatic luxation with dislocation of the connection between incus (1) and malleus (2) is seen. The typical icecone configuration of the…



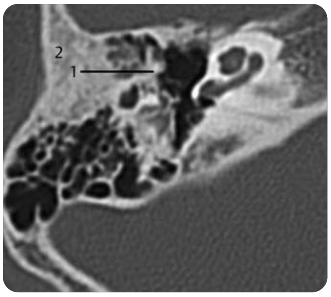
Axial bone CT in a 14-year-old following severe head injury shows a medial transverse, otic capsule-violating fracture traversing the vestibule, with…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 968.

MIDDLE EAR & MASTOID · BONE DISEASE

Fixation of the Ossicular Chain

- As mentioned above in differential diagnosis, particular attention must be paid to patients with conductive hearing loss in combination with a normal appearing eardrum and aerated middle ear.
- In case of ankylosis of the ossicular chain, accurate evaluation of thin axial and coronal slices may reveal deformities and/or fixation.



CT, axial. Although the conductive hearing loss might be ascribed to the atresia, the handle of the malleus (1) is fixed to the lateral bony wall of the…